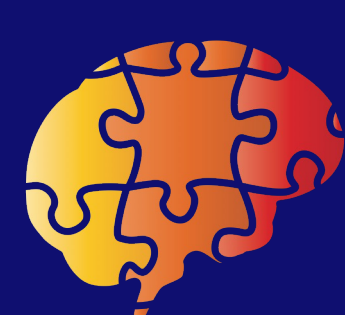
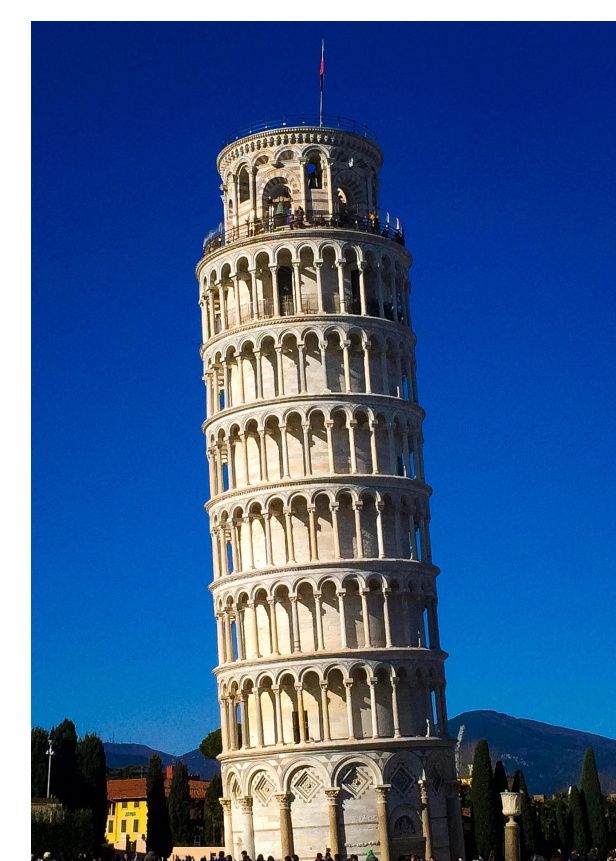
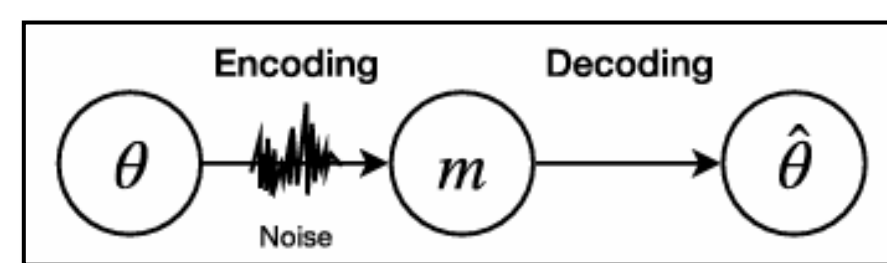




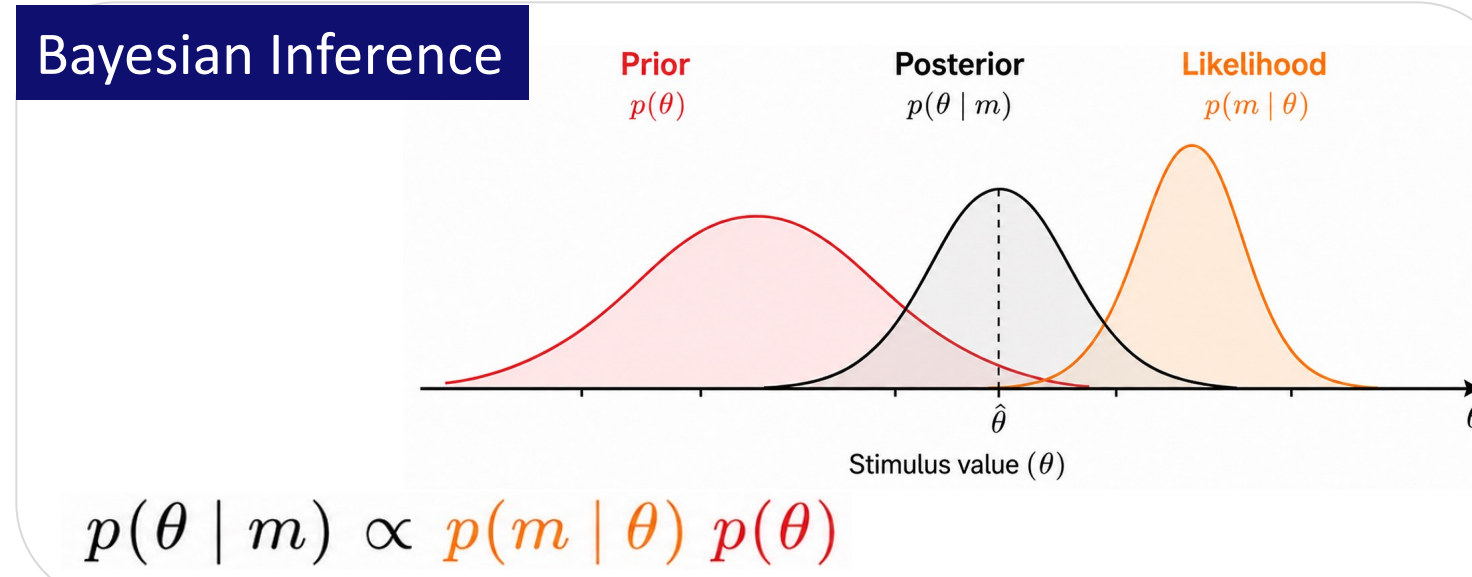
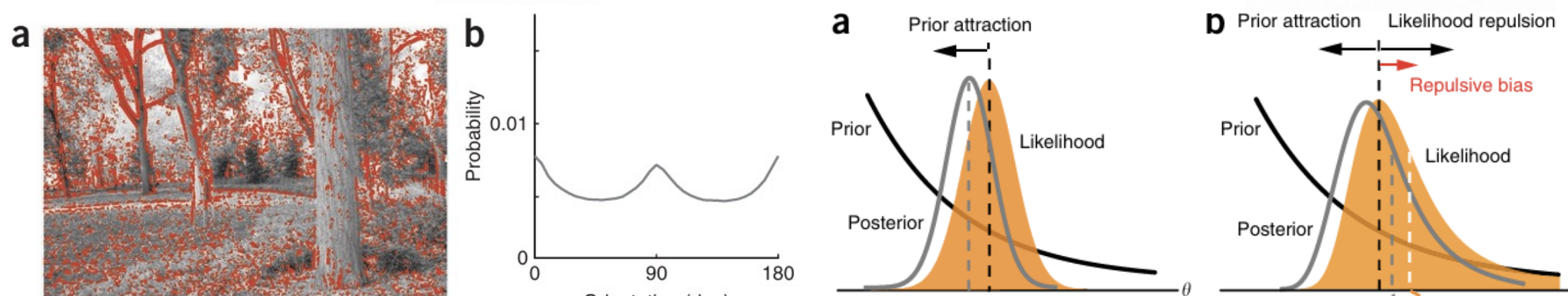
Background



“Leaning” Tower of Pisa



Bayesian Inference

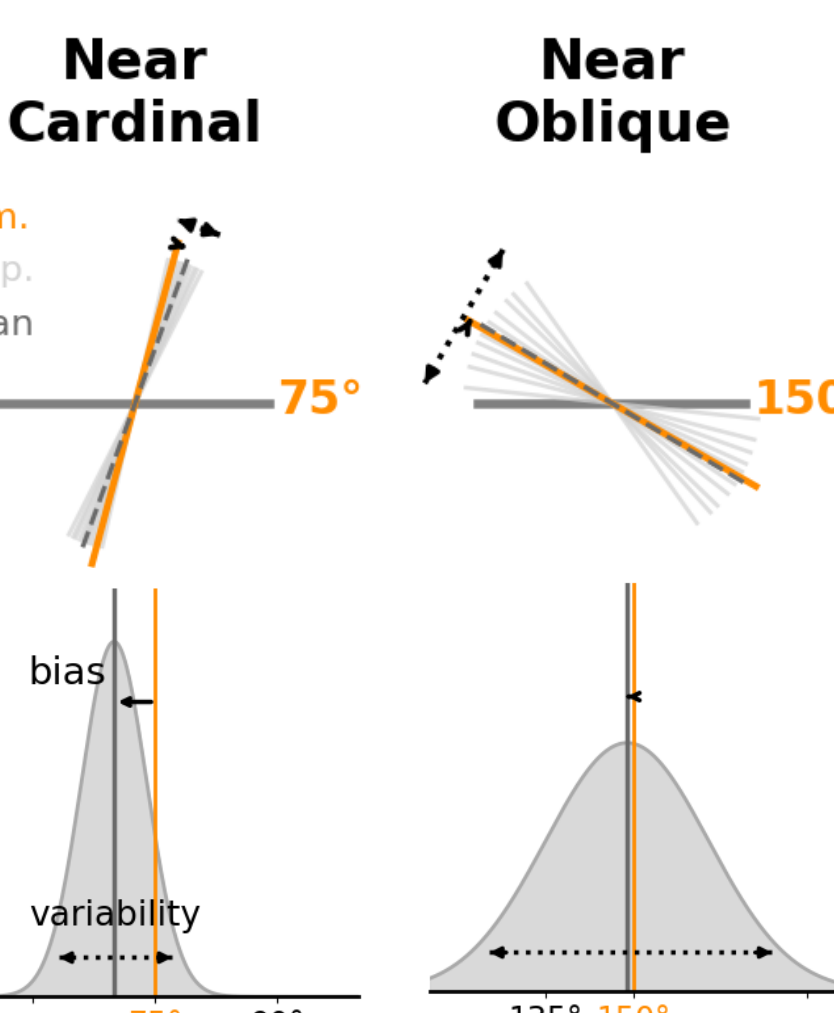
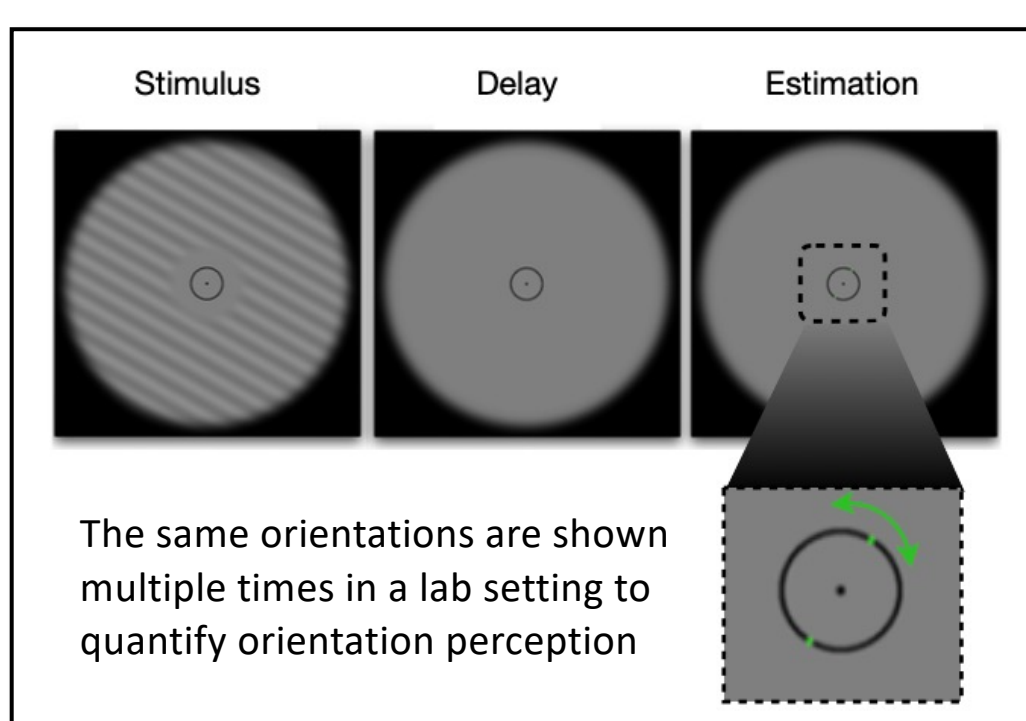
Prior $p(\theta)$ Likelihood $p(m|\theta)$ 

Girshick et al. (2011)

Wei & Stocker (2015)

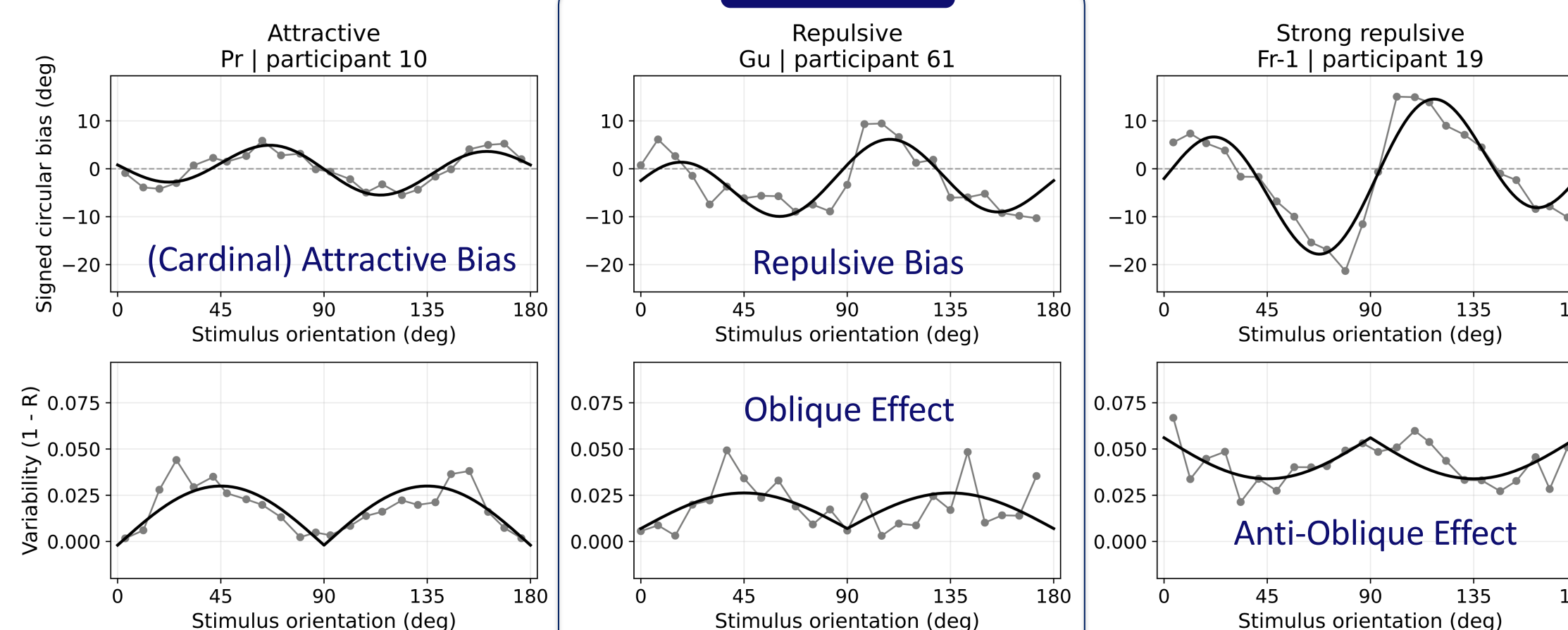
Orientation Estimation Task Analysis

Orientation Estimation Task



Bias & Variability

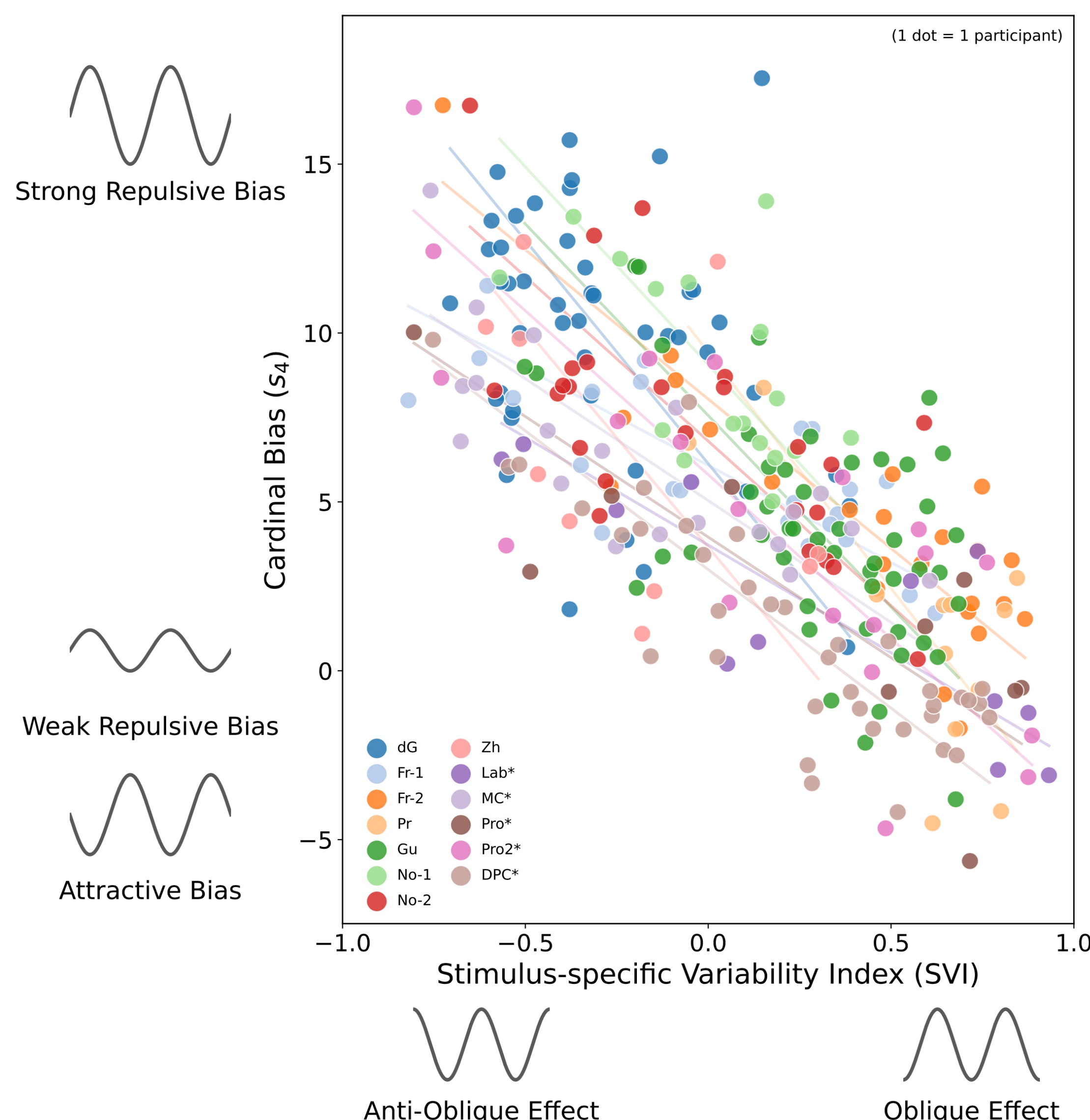
Classic



- Bias: How estimates deviate from the true stimulus on average
- Variability: How widely the estimates are distributed around their average

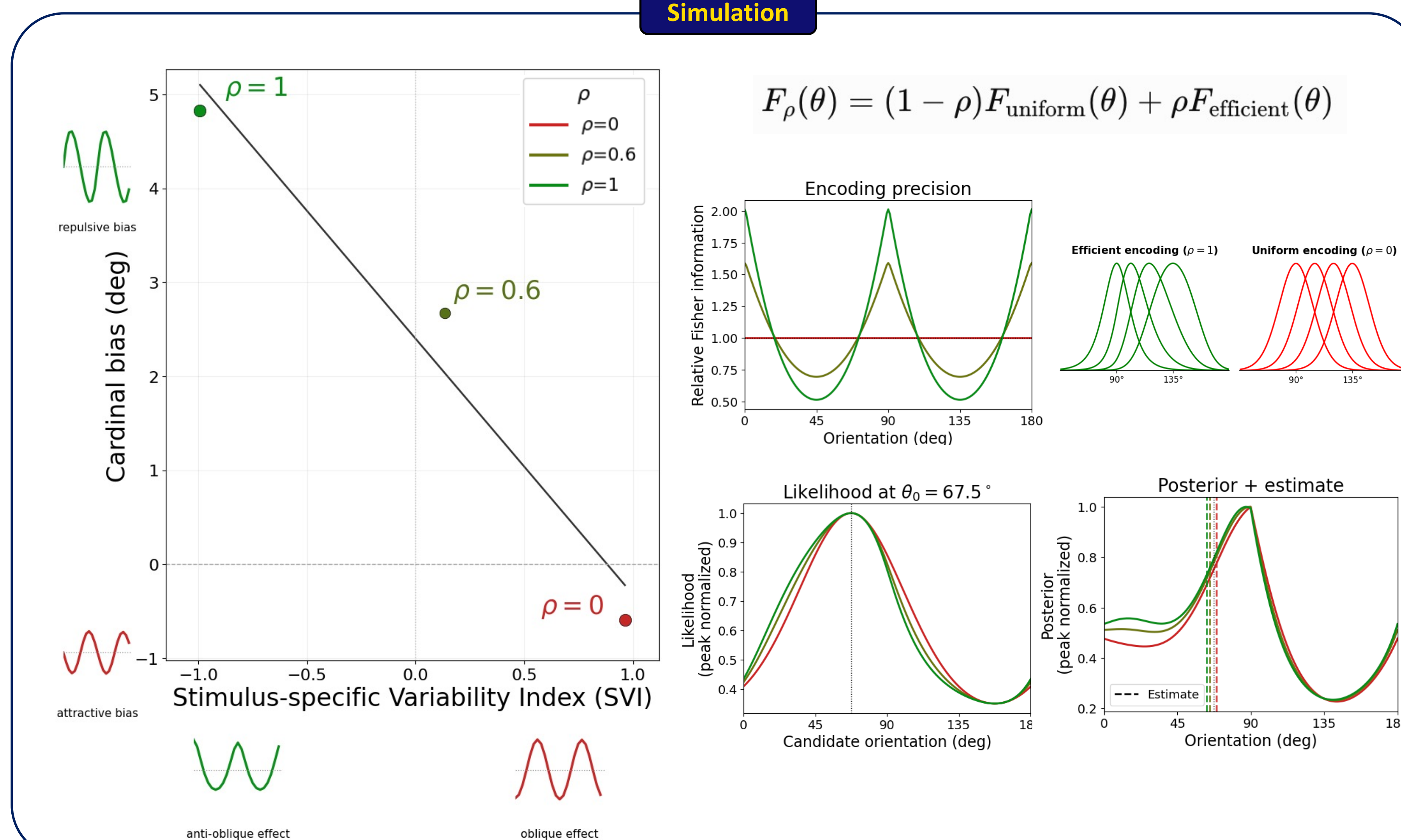
Results

Result 1. Individual differences in bias and variability follow a systematic relationship



Simulation

$$F_{\rho}(\theta) = (1 - \rho)F_{\text{uniform}}(\theta) + \rho F_{\text{efficient}}(\theta)$$



Result 2. Sensory uncertainty modulates orientation bias in an observer-specific manner

13 datasets/cohorts from published and in-lab studies

dG	de Gardelle et al. (2010), <i>Journal of Vision</i>
Fr-1, Fr-2	Fritsche et al. (2017), <i>Current Biology</i> , separate experiments
Pr	Pratte et al. (2017), <i>Journal of Experimental Psychology</i>
Gu	Gu et al. (2025), <i>Neuron</i>
No-1, No-2	Noel et al. (2021), <i>PLoS Biology</i> , ASD and TD cohorts
Zh	Zhang et al. (2025), <i>PNAS</i>

* Lab, MC, Pro, Pro2, and DPC were collected within our lab

Y axis: Cardinal Bias (s_4)

$$b(\theta) = \beta_0 + c_2 \cos 2\theta + s_2 \sin 2\theta + c_4 \cos 4\theta + s_4 \sin 4\theta$$

$$s_4 < 0: \text{attractive} \quad | \quad s_4 > 0: \text{repulsive}$$

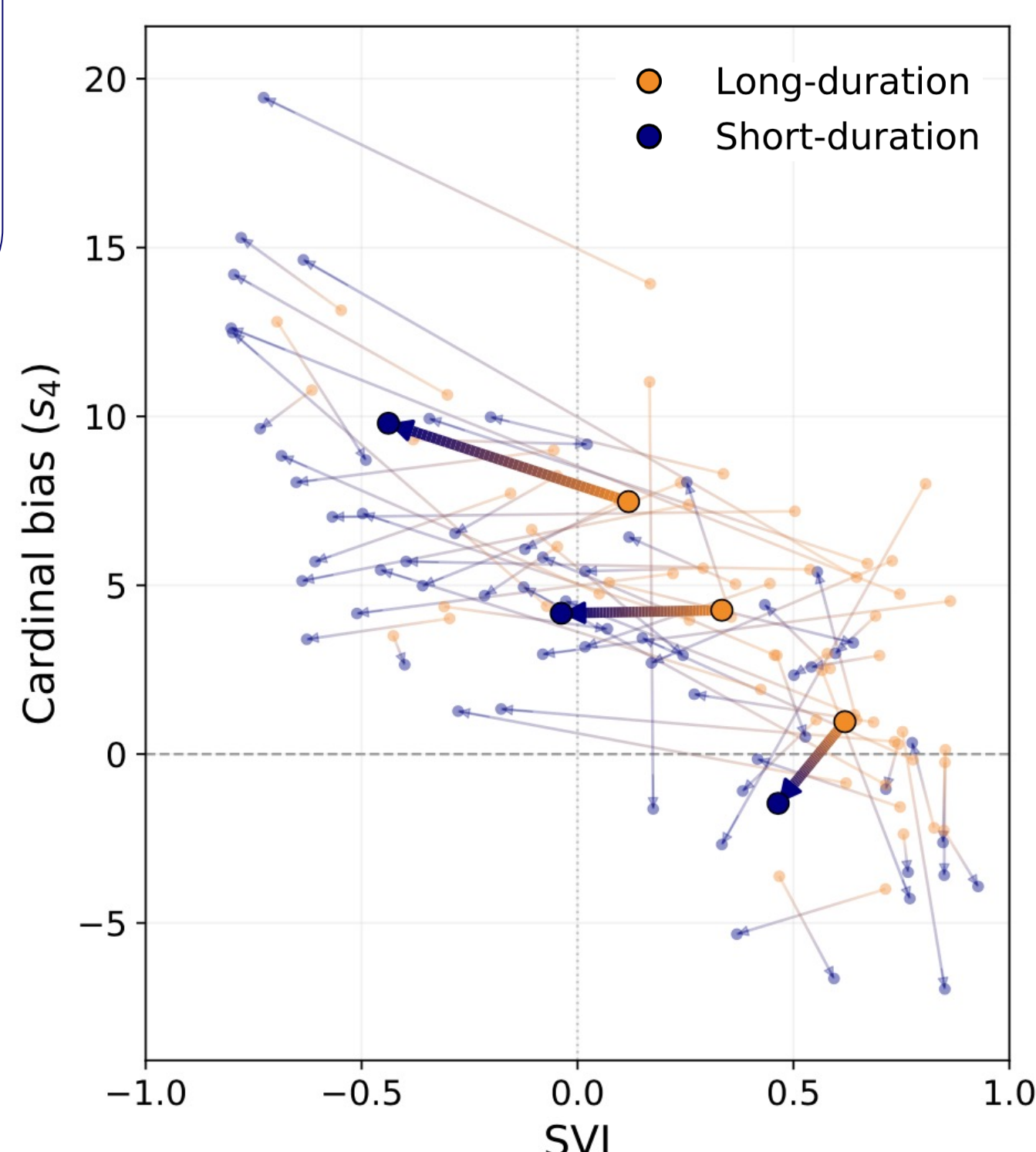
X axis: Stimulus-specific Variability Index (SVI)

$$R(\theta) = \left| \frac{1}{N} \sum_j e^{i2e_j} \right|, \quad V(\theta) = 1 - R(\theta)$$

$$\text{SVI} = r[V(\theta), |\sin 2\theta|]$$

$$\text{SVI} < 0: \text{anti-oblique} \quad | \quad \text{SVI} > 0: \text{oblique}$$

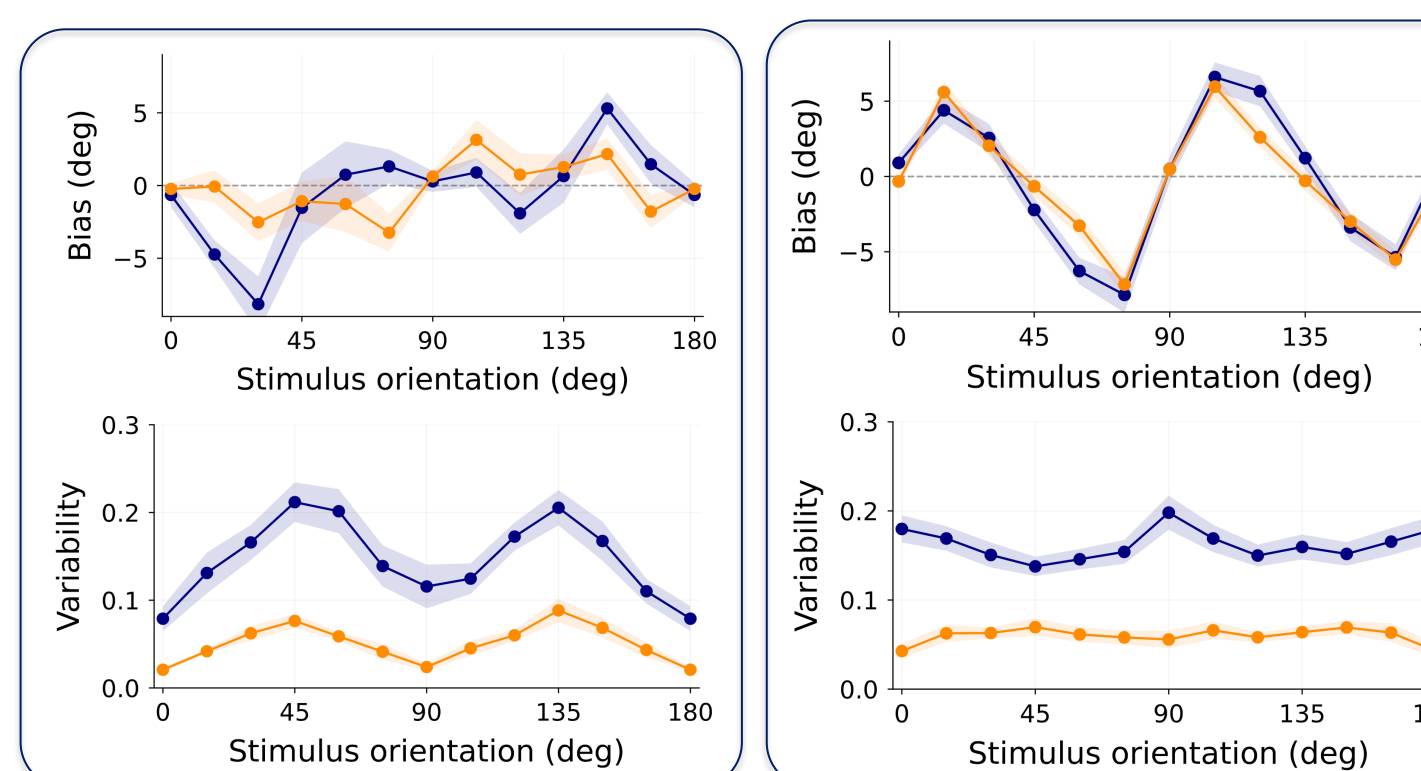
Uncertainty Manipulation Main Result



4 datasets with within-observer duration manipulation

Dataset	N	Target duration (ms)
1 Online prolific	Pro N=10	34 / 167 / 500
2 Offline lab exp	Lab N=12	50 / 500
3 Graduate Class	MC N=20	50 / 500
4 Online prolific 2 nd	Pro2 N=20	75 / 500

Attraction Group & Repulsion Group Profile Changes: Short vs Long



Simulation

